MILITARY SPECIFICATION

SODIUM OXACILLIN CAPSULES, 48s

1. SCOPE

1.1 This specification covers Sodium Oxacillin Capsules, Equivalent to 0.25 Gm of Oxacillin in each capsule. These capsules shall have a potency period of 24 months.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

U-C-115	Capsule, Gelatin.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-636	Box, Fiberboard.
PPP-B-665	Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676	Boxes, Set-Up, Paperboard.
PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary.

(Application for copies should be addressed to the J.B. Lippincott Company, Philadelphia, Pa.)

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
General Regulations for its Enforcement.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D.C. 20204.)

U.S. PHARMACOPOEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa.)

3. REQUIREMENTS

3.1 Material. Shall be capsules containing 250 mg of Oxacillin (5-methyl-3-phenyl-4-isoxazolyl penicillin) as the sodium salt, per capsule, within the designated assay limits for the capsule.

3.2 Certification. The finished capsules shall conform to the pertinent provisions of the General Regulations for the Certification of Antibiotics and Antibiotic-Containing Drugs, as promugated by the Food and Drug Administration (F.D.A.).

3.2.1 Each lot of the finished capsules shall be certified by the F.D.A.

3.3 Assay. The finished capsules shall assay to contain not less than 90.0 percent of the required amount of sodium oxacillin, when assayed as described in 4.4.2.

3.4 Moisture content. Moisture content in the finished capsules shall be not more than 6.0 percent.

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3.5 Ingredients.

3.5.1 Sodium oxacillin. The sodium oxacillin, used in the manufacture of the capsules, shall conform to the pertinent provisions of the General Regulations for the Certification of Antibiotics and Antibiotic-Containing Drugs.

3.5.2 Other ingredients. All other ingredients entering into the manufacture of the finished capsules, shall be of U.S.P. or N.F. quality or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.5.3 Identification. The ratio of the absorbancies A_{232} mu/A_{282} mu of the sample shall be basically identical with the ratio obtained from the standard, when determined as described in 4.4.1.

3.6 Capsules. The empty capsules used in the manufacture of the finished sodium oxacillin capsules shall conform to the applicable paragraphs of U-C-115, except as specified herein:

3.1 Delete in its entirety and substitute:

"3.1 Material. The gelatin, used in the manufacture of the pharmaceutical capsules, shall be type A or type B, or a suitable mixture of both and shall be in accordance with the requirements of the U.S.P. The gelatin may not contain more than 0.15 percent sulfur dioxide and may have a lower gel strength."

"3.2.3.3 Appearance and color. Shall be conventional-shaped capsules. The base and/or cap end of the capsule may be essentially clear and colorless, or may be colored. Uniformity in color, size, shape, and appearance shall exist amongst the finished capsules. The manufacturer's trademark or imprint may appear on the base and/or the cap of each capsule."

3.7 Stability. Capsules shall be so compounded as to provide physical stability of capsules and chemical stability of the ingredients. Capsules shall comply with all applicable requirements for the full potency period under normal storage and handling.

3.8 Filled capsules. The filled capsules shall be uniform and free from manufacturing or other defects, such as cracks, dents, splits, specks, etc.

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3.9 Odor. Capsules shall be odorless if they do not contain ingredients in which an odor is characteristic of the material. Capsules shall have no foreign odor or odor resulting from decomposition or deterioration.

3.10 Weight variation. The filled capsules shall conform to the U.S.P. "Weight Variation" test for capsules.

3.11 Net contents and variations in count. The number (count) of capsules in the immediate containers shall average not less than that quantity specified (48). No immediate container shall be more than 1 capsule short of the required amount.

3.12 Delivery. Not less than 22 months of the expiration dating period (potency period) shall remain at time of delivery to the Government.

3.13 Workmanship. The material and its containers shall be free from any defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Lot. For purpose of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of finished capsules produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major A	II	1.0	Filled Capsule
Major B	S-4	1.0	Filled Bottle
Minor	II	2.5	Filled Capsule

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4.2.1 For end item testing. Noncompliance with any of the following is cause for rejection.

Characteristic	Requirement paragraph	Inspection* level	Unit of product
Assay	3.3	S-1 (Composite from final filled bottle).	Filled Bottle
Identity	3.5.3		Filled Bottle
Net contents	3.11	S-2	Filled Bottle
Label adhesion	----	PPP-C-186 (10 minimum)	Filled Bottle
Bottle (style and design)	PPP-C-186	S-2	Empty Bottle
Closure (style and design)	PPP-C-186	S-2	Closure

* Where applicable, the same sample may be used for two (2) or more tests.

4.3 Examination. The sodium oxacillin capsules shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.2 and 4.2.1.

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4.3.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

Categories and defects*

Major A

101. Capsule not hard gelatin type, i.e., not type A, B, or a mixture of A and B.

102. Capsule not free of cracks, breaks, or splits.

103. Capsule not uniform in appearance.

104. Capsule not size suitable.

105. Capsule not free of foreign matter.

Major B

151. Capsule not free of foreign odor (other than characteristic odor).

152. Base and cap do not maintain tight friction closure (conventional type capsule).

153. Capsule not intact (separated).

154. Ingredients not free of dirt or foreign matter.

155. Desiccant not furnished.

156. Circular not furnished.

Minor

201. Capsule not free of surface blemishes.

202. Capsule not smooth.

*Examination is not restricted to the classified defects listed above.

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4.4 Tests.

4.4.1 Identity. The sodium oxacillin capsules shall be subjected to the following identity test:

Heat about 1/2 of the contents of one capsule in a Folin tube with 25 ml of 5N sodium hydroxide for 30 minutes in a boiling water bath. Cool rapidly, acidify 1 ml with 2 ml of 5N sulfuric acid, add 50 ml distilled water and extract with 50 ml of ethyl ether. Extract the ether with 50 ml 0.1N sodium hydroxide and dilute the alkaline extract to 100 ml with distilled water. Record the ultraviolet spectrum between 360 mu and 220 mu on a suitable spectrophotometer using 1-cm cells and 0.05N sodium hydroxide as a blank, and calculate the ratio A,232mu/A,282 mu. The spectrum shall be basically identical to that of a standard similarly treated.

4.4.2 Assay. The sodium oxacillin capsules shall be subjected to the following assay:

Reagent No. 1 - 0.01N Iodine Solution.

Weigh 2.54 Gm of resublimed iodine, chemically pure (C.P.), and 5 Gm of potassium iodide, C.P. into a small beaker, add 5 ml of distilled water, and dissolve with gentle stirring. Transfer to a 2-liter volumetric flask, dilute to volume with distilled water and mix well.

Reagent No. 2 - 0.01N Sodium Thiosulfate Solution.

Weigh 2.48 $\pm$ 0.01 Gm sodium thiosulfate pentahydrate, dissolve in 1 liter of distilled water in a volumetric flask and add 1 ml of chloroform as a stabilizer. Standardize against a standard of Sodium Oxacillin, as follows:

Transfer an accurately weighed 40 to 50 mg sample of sodium oxacillin to a 50-ml volumetric flask, dilute to volume with distilled water, mix, take 1.0-ml aliquots for assay, and proceed as described in this document under "Method", Run one standard with each series of potency determinations.

Standard Factor: $\frac{\text{mg standard} \times \text{standard potency (mcg/mg}}{\text{difference in titrations} \times 50 \times 1000}$

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Reagent No. 3 - Starch Indicator Solution.

Weigh 10 Gm of soluble starch and triturate into a smooth paste with 50 ml of cold distilled water. Pour into one liter of boiling distilled water, stir, and boil until only slightly opalescent. Store in a sterile bottle with 1 ml of metallic mercury as a preservative.

Transfer the contents of one capsule (accurately weighed by difference) to a blender, add 100 ml of distilled water and blend for 3 minutes. Add 150 ml of distilled water and blend for another minute. Filter through Whatman No. 2 filter paper, discard the first 10 ml of filtrate, transfer 1.0 ml of the subsequent filtrate to each of 2 flasks and proceed as directed as follows under "Method".

"Method"

Place the accurately measured amount of sample solution containing approximately 1 mg activity per ml in each of two 125-ml, glass-stoppered Erlenmeyer flasks. To the first (test) flask add 1.0 ml of 1N sodium hydroxide, mix well, and set aside for exactly 15 minutes. While the test flask is standing, add exactly 5.0 ml of Reagent No. 1 to the second (blank) flask, mix well and immediately titrate the excess iodine with Reagent No. 2, using two drops of Reagent No. 3 as internal indicator and record as "blank titration". After the test flask has stood the required 15 minutes, add 1.0 ml of 1.1N hydrochloric acid to neutralize the alkali and make the solution acid (pH about 2). Then add 5.0 ml of Reagent No. 1, mix well, stopper the flask and allow to stand exactly 15 minutes. At end of this time titrate the excess iodine, as in the case of the blank flask, and record as Test Titration. Determine the difference between the Blank and the Test.

Calculation: $\frac{\text{titration difference x standard factor x dilution}}{\text{Gm (or ml) sample}} =$

mg Oxacillin per Gm (or ml)

Not less than 225 mg per capsule (90.0 percent) shall be obtained.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified herein and shall be in accordance with all applicable paragraphs of PPP-C-186.

5.1.1 Immediate containers. Each immediate container shall comply with the following classification:

GROUP A	CLASS 1	TYPE e	STYLE 2	GRADE 2
CLOSURE J			SEAL C - Shall be an outer aluminum, roll-on tamperproof seal.	

5.1.2 Desiccant. A suitable desiccant, appropriately packaged and marked, shall be packaged in each immediate container.

5.2 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug, and Cosmetic Act, and shall include the information required below:

5.2.1 Immediate containers. Each immediate container label shall bear the following information. However, the information is not required to appear in the sequence indicated.

(a) the item identification designated as "SODIUM OXACILLIN CAPSULES"

(b) the quantity of active ingredient designated as "Equivalent to 0.25 Gm of Oxacillin in each capsule"

(c) the phrase "48 Capsules" or a similar phrase

(d) the stock number designated as "FSN 6505-226-1202" or "Stock No. 6505-226-1202"

(e) the lot or control number

(f) the expiration date

(g) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear.

When both names are placed on the label, the following designation shall precede the names:

"MFR" for the manufacturer and
"CONTR" for the contractor.

(See additional labeling on Page 10)

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(h) the statement "Caution: Federal law prohibits dispensing without prescription"

(i) the following or similar statement:

"Usual adult dose: 2 capsules (500 mg) every 4 to 6 hours"

5.2.1.1 Extended potency lines. Paragraph 3.9.8 of PPP-C-186 shall apply to the immediate container labels.

5.2.2 Unit packages. Each unit package label shall bear the same information as required for the label of the immediate container.

5.2.3 Circulars. A circular, brochure, or other printed matter shall be packaged within each unit package, setting forth as a minimum: Indications; Administration and dosage; and Precautions and contraindications.

5.3 Packaging.

5.3.1 Unit of issue. One (1) bottle, as specified, constitutes one unit of issue.

5.3.2 Unit package. Each unit shall be packaged as specified in 5.2.5.

5.3.3 Intermediate package. The number of units of issue specified in table I, column 2, for Procedure Code No. 3, indicated in column 1, shall be packaged in an intermediate package constructed in accordance with the applicable paragraph referenced in column 3, except that partitions shall not be required. As applicable, commercial colors shall be acceptable when carton conforming to PPP-B-566, PPP-B-665, or PPP-B-676 is utilized.

5.4 Packing. The number of units contained in the intermediate packages and total number of units of issue for the applicable procedure code, indicated in column 4, shall be overpacked in an exterior container, constructed in accordance with the applicable paragraph referenced in column 5 (level B), or column 6 (level A), for the level of protection specified in the procurement document. Bursting strength of carton shall be in accordance with special requirements in table I of PPP-B-636, in lieu of bursting strength specified in the applicable paragraph referenced in column 5 (level B). Method I closure of PPP-B-636 shall be utilized on level B fiberboard packs. In addition, in line 10 of 5.4.4, delete "IV" and substitute "III; or PPP-B-636, type CF, class weather-resistant"; at end of 5.4.4.1, add "Case liner shall not be required for fiberboard boxes." Add the following new paragraph: "5.4.4.1.1 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636."

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5.5 Marking.

5.5.1 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Lot (control) number, contract or purchase order number, and name of contractor shall be shown. The expiration date shall be shown in lieu of date packed.

5.5.2 Exterior containers. Exterior containers shall be marked in accordance with MIL-STD-129. Lot (control) number and expiration date (in lieu of date packed) shall be shown.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Level of packing (see 5.4).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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